Ge YAN

Address: UC San Diego, 9500 Gilman Drive, La Jolla, CA 92093-0021

Mobile: +1(858)531-8601 | Email: geyan@ucsd.edu

EDUCATION

School of Mathematical Sciences, Peking University Sep 2017-Jun 2021

B.S. in Information and Computing Science (GPA 3.484/4)

Core curriculum: Numerical Linear Algebra, Convex Optimization, Machine Learning, Algorithms for Big Data Analysis, Introduction to Numerical Analysis, Applied Stochastic Processes.

Department of Electrical and Computer Engineering, UC San Diego Sep 2021-Present

M.S. in Machine Learning and Data Science (EC93) (Current GPA 4.000/4)

Core curriculum: GPU Programming, Statistical Learning, Linear Algebra, Probability and Statistics for Data Science.

RESEARCH EXPERIENCES

Neural network architecture design with ordinary differential equation numerical schemes Apr 2019-Oct 2019

Advisor: Bin Dong, Associate Professor, Beijing International Center for Mathematical Research (BICMR), Peking University

- Implemented a Resnet containing three submodules and 18 layers with ReLU activate function using Pytorch
- Refined the skip connection in Resnet leveraging numerical algorithms in solving ordinary differential equations, like Adams algorithm and three-step linear multistep method
- Explored the influence of different skip connections inspired by different numerical algorithms to the model performance on CIFAR-10, a popular image classification dataset
- Achieved 92.2% accuracy on CIFAR-10 using the refined model, outperformed the original Resnet by 0.9%

Exploration on neural network similarity via white-box adversary attack Mar 2020-Jun 2020

Advisor: Liwei Wang, Professor, School of Electronic Engineering and Computer Sciences, Peking University

- Applied PGD attack algorithm to implement a white-box attack algorithm that casted attack on one model and the adversary accuracy on another one depended on the similarity between attacked network and target network
- Proposed the idea to use white-box attack to measure the similarity between features learned by differently initiated networks
- Trained 15 differently initialized Resnet models and used our method to compare the similarity between these models
- Collaborated with teammates on all the codes with Git

Failure probability estimation via Bayesian neural networks Mar 2021-Jun 2021

Advisor: Chao Yang, Professor, School of Mathematical Sciences, Peking University

- Introduced Bayesian neural networks as surrogate model in surrogate method for Failure probability estimation.
- Applied importance sampling implemented with generative networks to Monte Carlo procedure of the problem.
- Experimented the method on a fluid flow problem and compared result with plain Monte Carlo method. Achieved similar accuracy with approximately 1% of sampling numbers of plain Monte Carlo.

SKILLS

Language: English (TOFEL: 105), GRE: Analytic Writing: 4.0 Verbal: 159 Quantitative: 170

Programming with Python, Matlab, C and cuda C

Familiarity with python libraries including Pytorch, Numpy, Pandas, etc.

Experience on training various network models on Linux server.